

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and non-functional requirements.

1. System Requirements

System Name: LibraryManagementSystem

Architecture Style: Microservices Architecture

Functional Requirements

- The system must detect the user's face using the YOLOv5 model.
- The system must detect the user's hand movement using Background Subtraction (BS).
- The system must recognize gestures.
- The system must recognize faces.
- The system must track hands.
- The system must support single-hand interaction.
- The system must allow users to customize actions for specific gestures.

Non-Functional Requirements

- Competitive price.
- Real-time collaboration.
- Low-latency.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, connectors, and constraints.

```
System LibraryManagementSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices Architecture";
  Property source : String = "AI";
  Component FaceDetectionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Detect the user's face using the YOLOv5 model, and use contour tracking for face detection";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component HandMovementDetectionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Detect the user's hand movement using Background Subtraction (BS)";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
}
```

```

}
Component GestureRecognitionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Recognize gestures using contours detection, Gaussian blur,
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component FaceRecognitionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Recognize faces using contours detection, Gaussian blur, Background
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component HandTrackingModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Track hands using contours detection, Gaussian blur, Background
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component Single-HandInteractionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Support single-hand interaction using contours detection, Gaussian
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component CustomizableGestureModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Allow users to customize actions for specific gestures using
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment FaceDetectionModule.out to FaceRecognitionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment HandMovementDetectionModule.out to HandTrackingModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment HandMovementDetectionModule.out to GestureRecognitionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment GestureRecognitionModule.out to Single-HandInteractionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment GestureRecognitionModule.out to CustomizableGestureModule.in {
    Property delivery : String = "at-least-once";

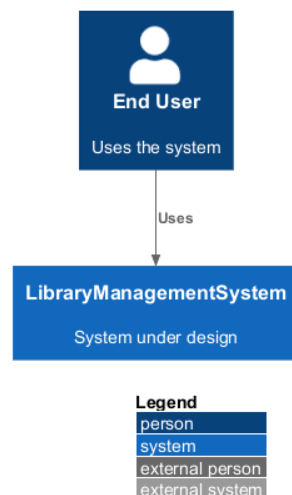
```

```

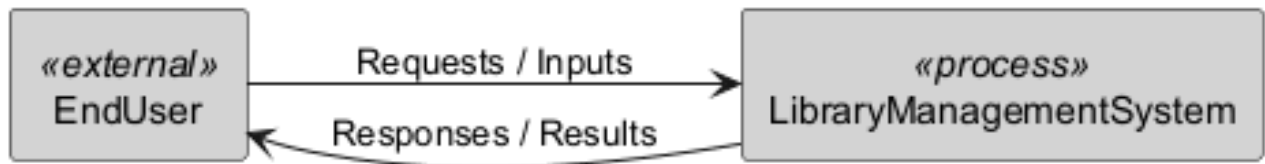
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment HandTrackingModule.out to Single-HandInteractionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CustomizableGestureModule.out to Single-HandInteractionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

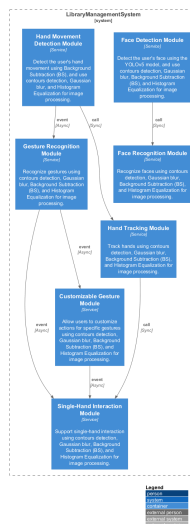
3. Context View



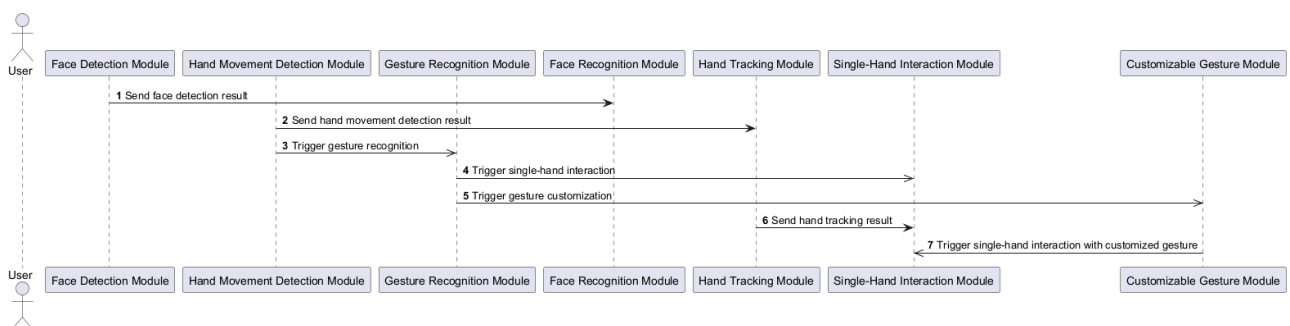
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

